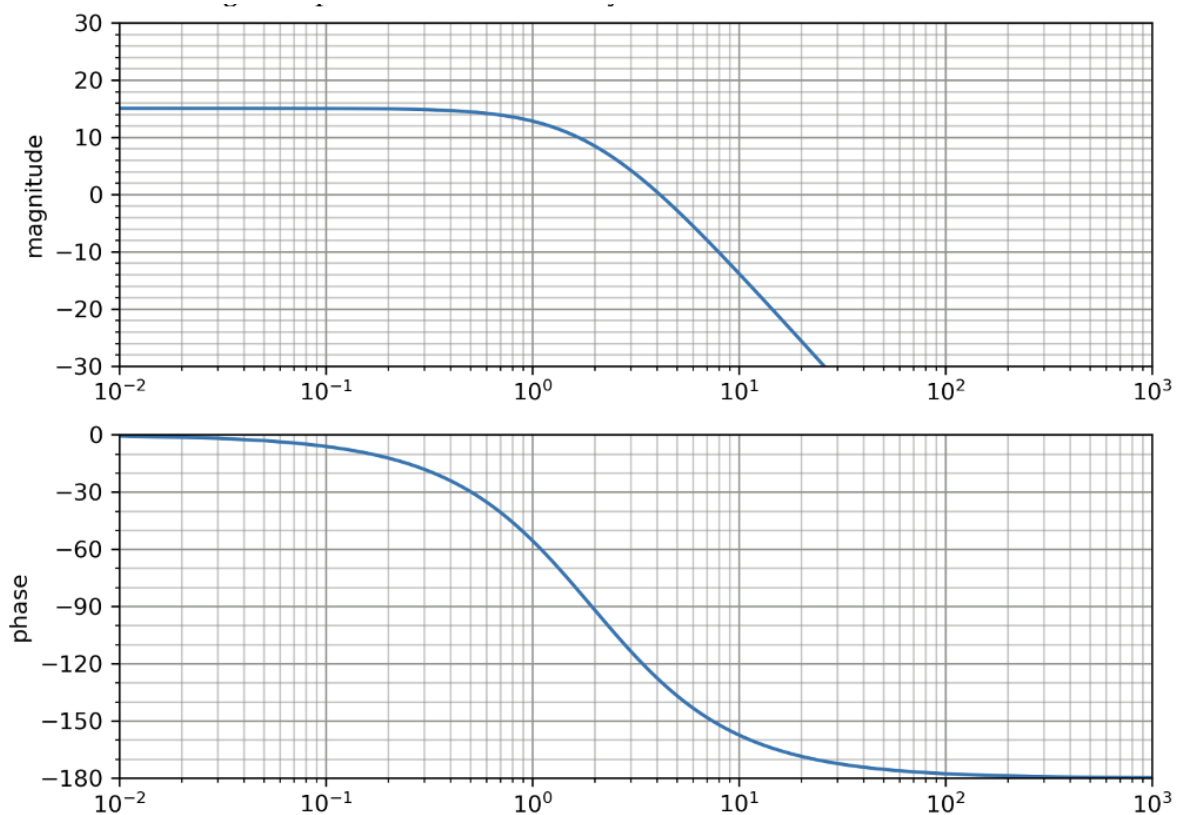


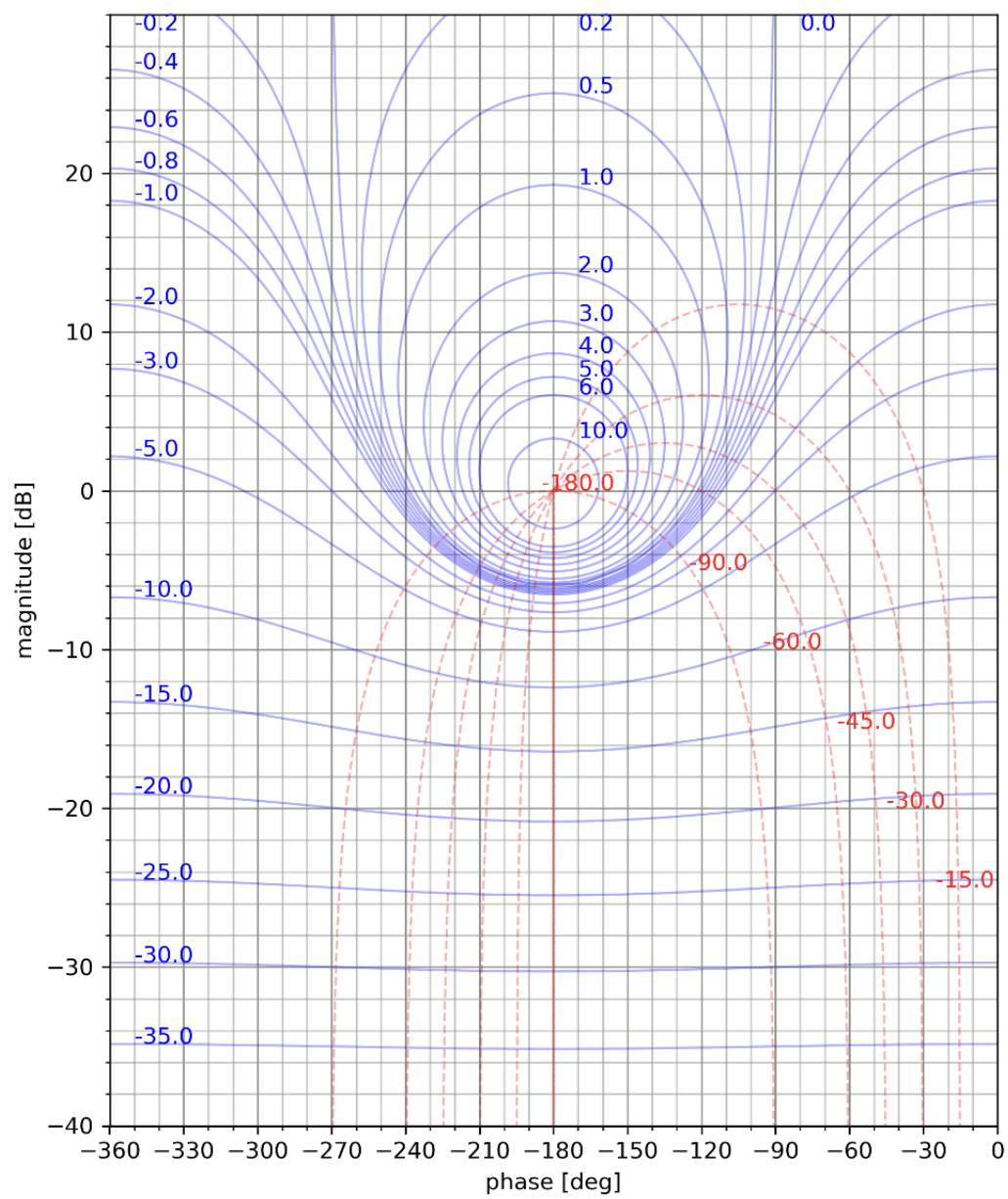
University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering
Problems Set 6.1

Question 1

Consider the following Bode plot of a second-order system:



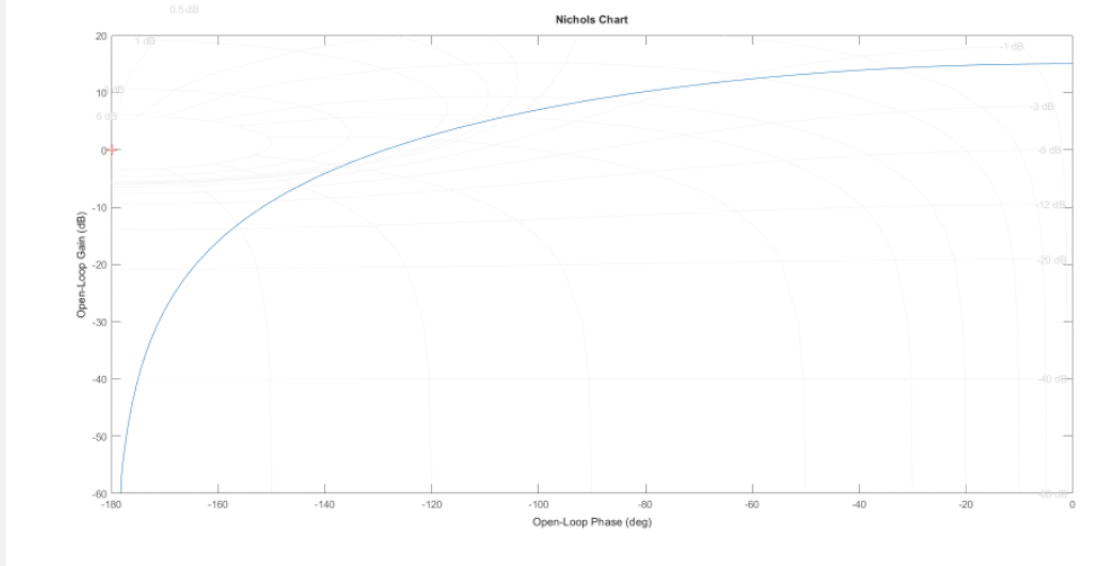
- A) Based on the Bode plot, sketch a rough Nichols chart on the template provided. Label the points corresponding to the frequencies $\omega = [0.1, 0.5, 1, 5, 10]$
- B) Using the Bode and/or Nichols plots, approximate the (i) position error constant, (ii) damping ratio, (iii) time constant and (iv) pole locations for the closed-loop system.



Solution:

Check: the open-loop transfer function used to generate the Bode plot is $G(s) = \frac{21.25}{s^2 + 4s + 3.75}$

a) Plotting the Nichols chart for this system using MATLAB, we obtain:



b)

i) The transfer function for the closed-loop error is given by $G_e = \frac{1}{1+G(s)}$, so the position error constant will be given by the final value of this transfer function for a step input, as shown below. From the Bode plot, the open loop gain of the system is around 15dB or 5.62 in linear units.

$$e(\infty) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \cdot \frac{1}{1+G(s)} = \frac{1}{1+5.62} = 0.15$$

ii) The damping ratio can be approximated by the phase margin with $\zeta = \frac{\phi_m}{100}$. From the Bode plot, the gain crossover occurs at 4 rad/s. At this frequency, the phase is -130°, so the phase margin is 50° and the damping ratio is 0.5.

iii) From the previous question, the system has complex poles, so we must find the natural frequency to determine the time constant. On the Nichols chart, this will be the frequency at which the system intersects the largest M-circle, i.e. where a resonant peak occurs. In this case, the system intersects the 1 dB M-circle at $\omega \approx 4$ rad/s.

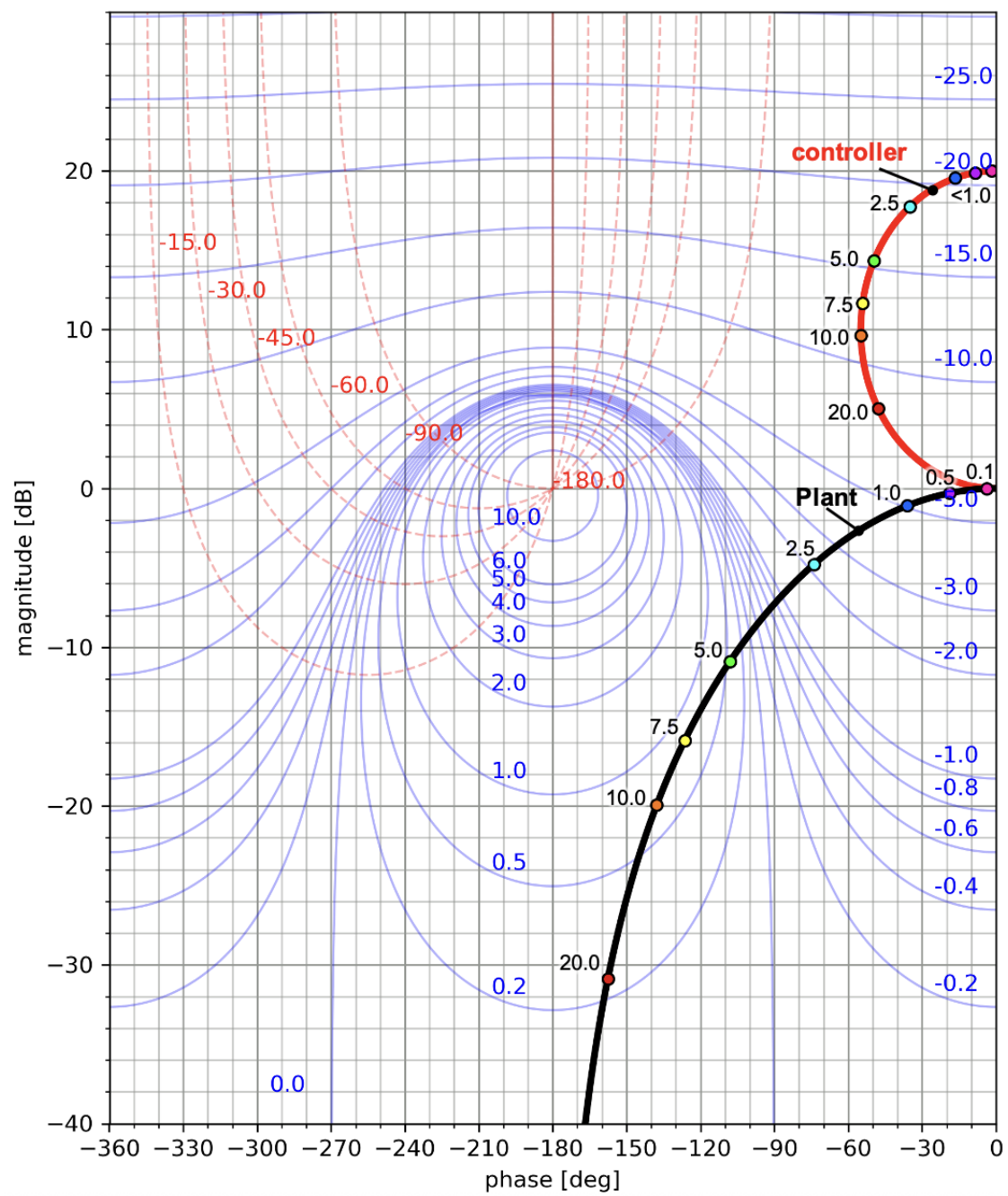
The time constant is given by the real part of the poles: $\tau = \frac{1}{\zeta\omega_n} = \frac{1}{(0.5)(4)} = 0.5$.

iv) The poles for an underdamped second-order system are located at $p = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -2 \pm j3.46$.

Question 2

The inverse Nichols chart on the next page shows the frequency responses of a system and a proposed controller for the system.

- A) Roughly sketch the combined response of the controller and plant on the inverse Nichols chart.
- B) Would you recommend using this compensator? Explain.
- C) If you decide to use a proportional controller instead, what is the largest gain that can be applied without
 - i. destabilising the system?
 - ii. violating the constraint: $\left| \frac{1}{1+L(s)} \right| \leq 2$ for all ω (note: the constraint is not posed in dB)
- D) What is the sensitivity of the (uncompensated) system at a frequency of 5 radians per second? If the open-loop system experiences a disturbance at this frequency, would you expect the effect of the disturbance to be larger or smaller in the output of the closed-loop system?

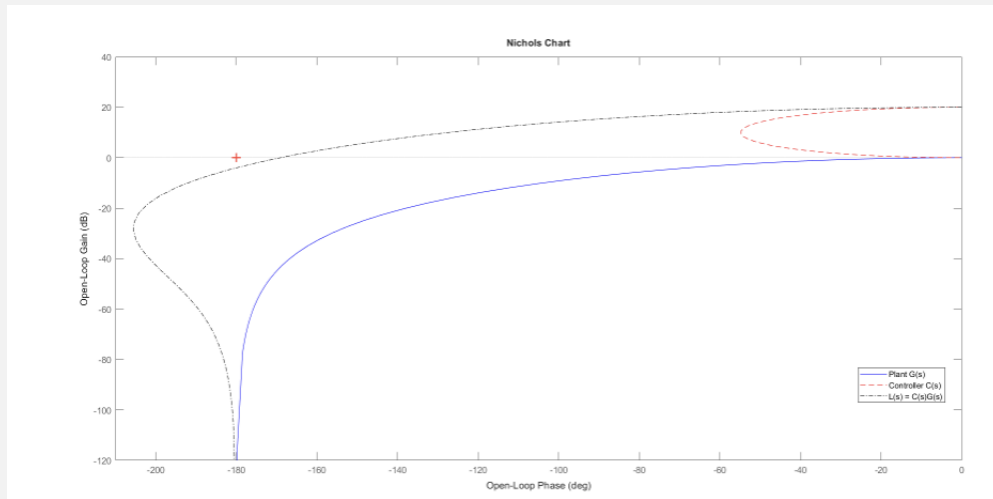


Solution:

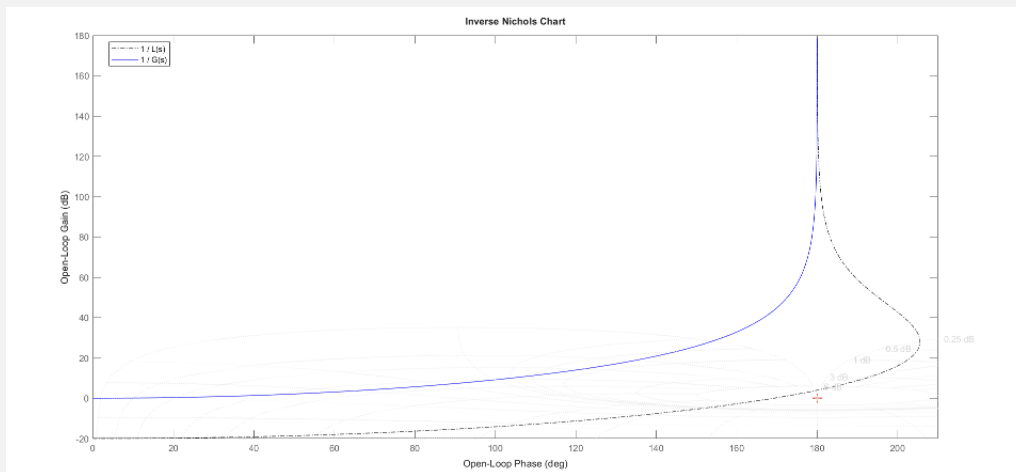
Check:

- Plant $G(s) = \frac{12}{(s+2)(s+6)}$
- Controller is a lag compensator $C(s) = \frac{s+30}{s+3}$

a) The combined open-loop transfer function yields the following Nichols chart:



Since the M-circles can't be flipped in MATLAB, the inverse Nichols Chart can be obtained by flipping the Nichols chart above:



b) No, because the compensator will reduce the system's gain and phase margin from being infinite to 4 dB and 10°, respectively, which is very small. Moreover, where the original system had a sensitivity of less than 1 dB over all frequencies, the compensated system has a sensitivity of greater than 6 dB between 5 and 8 rad/s, making it very susceptible to disturbances, particularly those which lie in that frequency range. As such, the compensator will make a fairly robust system into a very sensitive one.

c)

i. Since the gain margin of $G(s)$ is infinite, the gain can be arbitrarily large without destabilising the system.

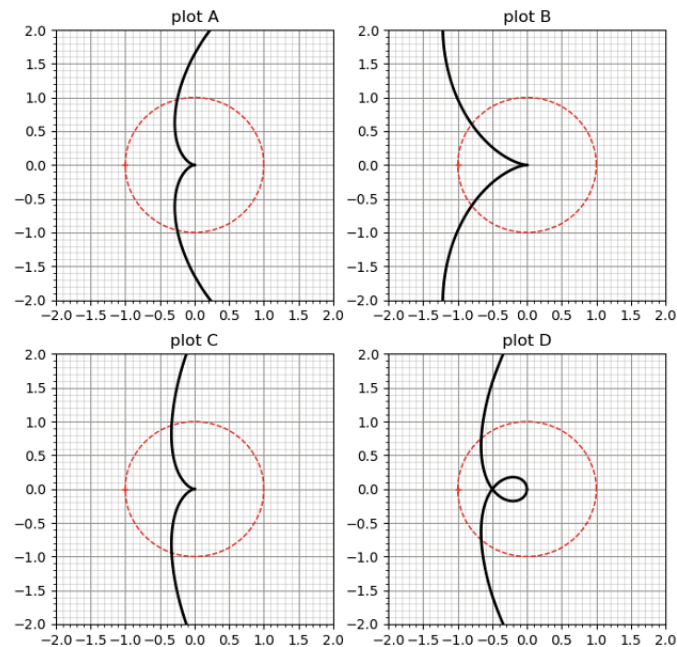
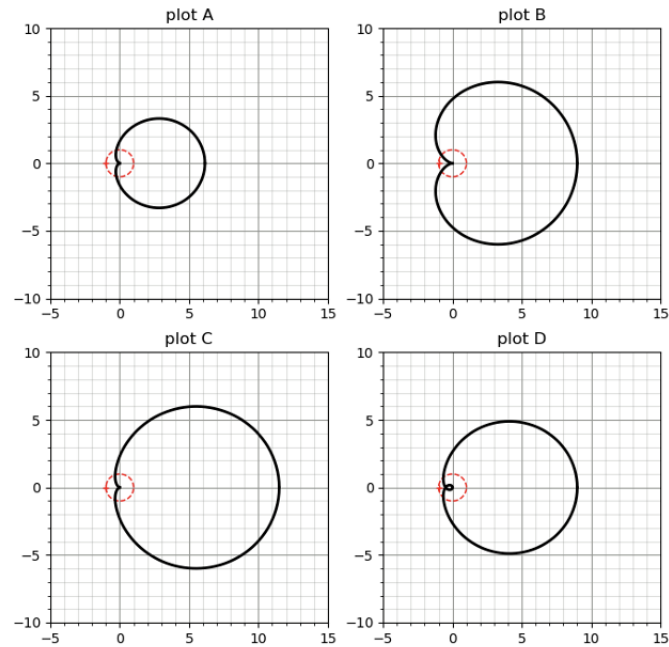
ii. The sensitivity must be less than 6 dB for all frequencies. The closest point to this M-circle on the Inverse Nichols chart can be shifted up by $26 - 2 = 24$ dB before this occurs, so 24 dB.

d) At this frequency, the Inverse Nichols chart intersects the 0.5 dB M-circle, so this is the sensitivity at 5 rad/s. Hence, disturbances at this frequency will be amplified, meaning their effect will be larger in the output of the closed-loop system.

Question 3

One of the four Nyquist plots in the following figure shows a system with infinite gain margin, a phase margin of > 60 degrees and a closed-loop position error of $< 10\%$

For each plot, state whether it has these characteristics. If not, explain why you are excluding it.



Detailed View of Nyquist Plots

Solution:

Plots A, B and C have infinite gain margin, since they don't intersect the negative real axis, meaning no amount of gain will cause the plot to encircle -1.

The phase margin is how much the points on the Nyquist plot that lie on the unit circle must rotate by to reach -1. By inspection, plots A, C, and D have a phase margin greater than 60 degrees since they intersect the unit circle at points close to 90 degrees.

Using the final value theorem, the position error constant is given by:

$$e(\infty) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \cdot \frac{1}{1 + G(s)} = \frac{1}{1 + A} < 0.1$$

Hence, to achieve a position error constant of less than 0.1, the gain must be greater than 9. The open-loop gain of the system is the point where the Nyquist plot intersects the positive real axis. Plot C is the only plot that intersects the positive real axis past 9, hence it is the only system that meets this requirement.

Question 4

Two lead compensator designs are proposed for the system $G(s) = \frac{1}{(s+3)(s+1)}$:

$$C_1(s) = \frac{K(0.09523s + 1)}{(0.00396s + 1)}$$

$$C_2(s) = \frac{K(0.1s + 1)}{(0.01091s + 1)}$$

The final system must have a percentage overshoot of $< 10\%$ and settle in $< 60\%$ of the fastest time achievable using proportional control.

- A) For each suggested design, identify the minimum gain K that would satisfy this requirement. What will the position error constant for each design be at this gain? (Hint: find a pole location that *just* satisfies these requirements first)
- B) Find the maximum phase lead for each compensator, and the frequency where it occurs.
- C) Which design would you recommend implementing? Explain.

Question 5

Consider the unstable system $G(s) = \frac{1}{(s+2)(s-0.5)}$

You must design a lead compensator to stabilise this system in closed loop. The compensated system must settle to within 5% of its final value in < 3 seconds and have a damping ratio of > 0.7 .

- A) Draw a root locus for the system and mark the region of the s-plane representing pole positions that satisfy the specifications.
- B) You decide you want the closed-loop poles to be the complex pair $-3.2 \pm j3$. Find the angles of the open-loop poles to the position $s = -3.2 + j3$ and calculate the angle the compensator must contribute to complete the angle criterion at this point.
- C) Now you need to place the zero for the compensator:
 - i. Technically, it is possible to satisfy the angle criterion at $s = -3.2 + j3$ using a zero at -2.1. Find the compensator pole position that corresponds to this zero and explain why this compensator would not be advisable.
 - ii. Explain why the zero should not be placed further to the left than -12.47
- D) You decide to try placing the zero at -10.
 - i. Calculate the required compensator pole position.
 - ii. Write an expression for your compensator in the form, $C(s) = \frac{K(\frac{1}{z}s+1)}{\frac{1}{p}s+1}$, where z is the zero position, p is the pole position, and K is the gain required to place the closed-loop poles at the target positions (Calculate K).
 - iii. If the system is required to track a position input with $> 90\%$ accuracy, is this an acceptable design? (Hint: don't forget to take the gain of the uncompensated plant into account.)

Question 6

Design a lead compensator to allow the spring mass damper system.

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

with output position $x(t)$, input force $F(t)$, and parameter values $m = 2$, $b = 5$, $k = 10$
To satisfy the following performance requirements in closed loop:

- Position error constant $< 5\%$
- Damping ratio of > 0.4
- Settling time to 5% of the final value of < 1 second

Your design should have a phase margin of at least 60% to allow for sampling effects.